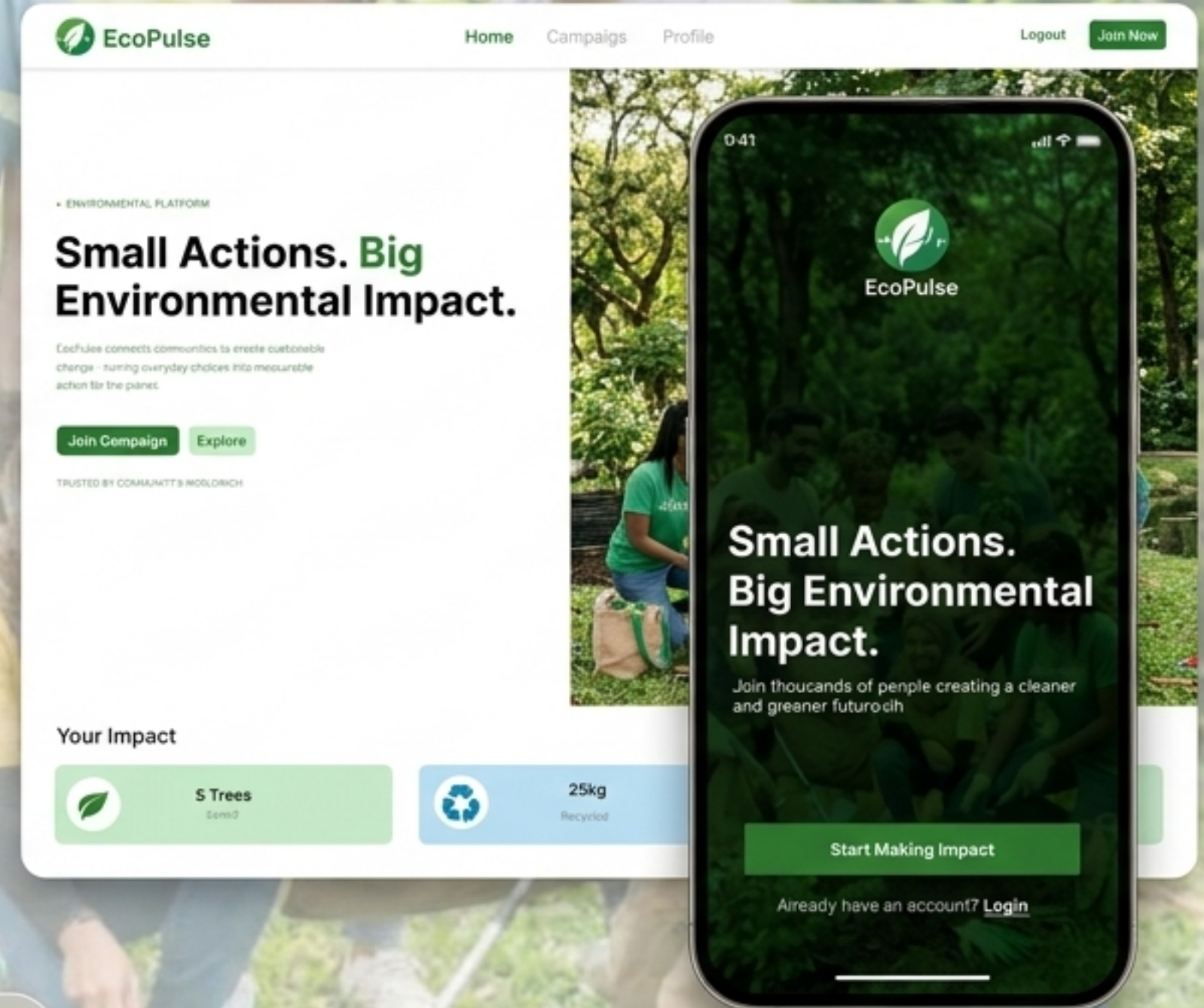


EcoPulse

A UI/UX Case Study in Environmental Engagement



Designed by Mohamed Mohamed Hussein | Human-Computer Interaction

The Engagement Gap



People want to help the environment but are blocked by poor design.

Current platforms suffer from cluttered information, confusing navigation, and a lack of engaging feedback.

The EcoPulse Mission



An intuitive digital platform that removes friction.

Objective: Make discovering and joining local eco-activities effortless, track personal impact, and motivate users through gamification across all devices.

User Ecosystem & Cognitive Design

Attention Span



Solution: Large bold headlines ('Small Actions. Big Environmental Impact.') and chunked UI to prevent cognitive overload.

Memory Constraints



Solution: Persistent details on campaign screens to eliminate recall effort.

Perception & Trust



Solution: Primary Dark Green palette applied to leverage psychological associations with nature and safety. High contrast text.

Motor Skills



Solution: Expanded, thumb-friendly tap targets, specifically on the mobile interface.



Meet Ahmad Farhan

24, University Student, Kuala Lumpur

📱 **Tech Literacy:** High (Smartphone Primary)

**I want to help the environment,
but I need something easy to use.**

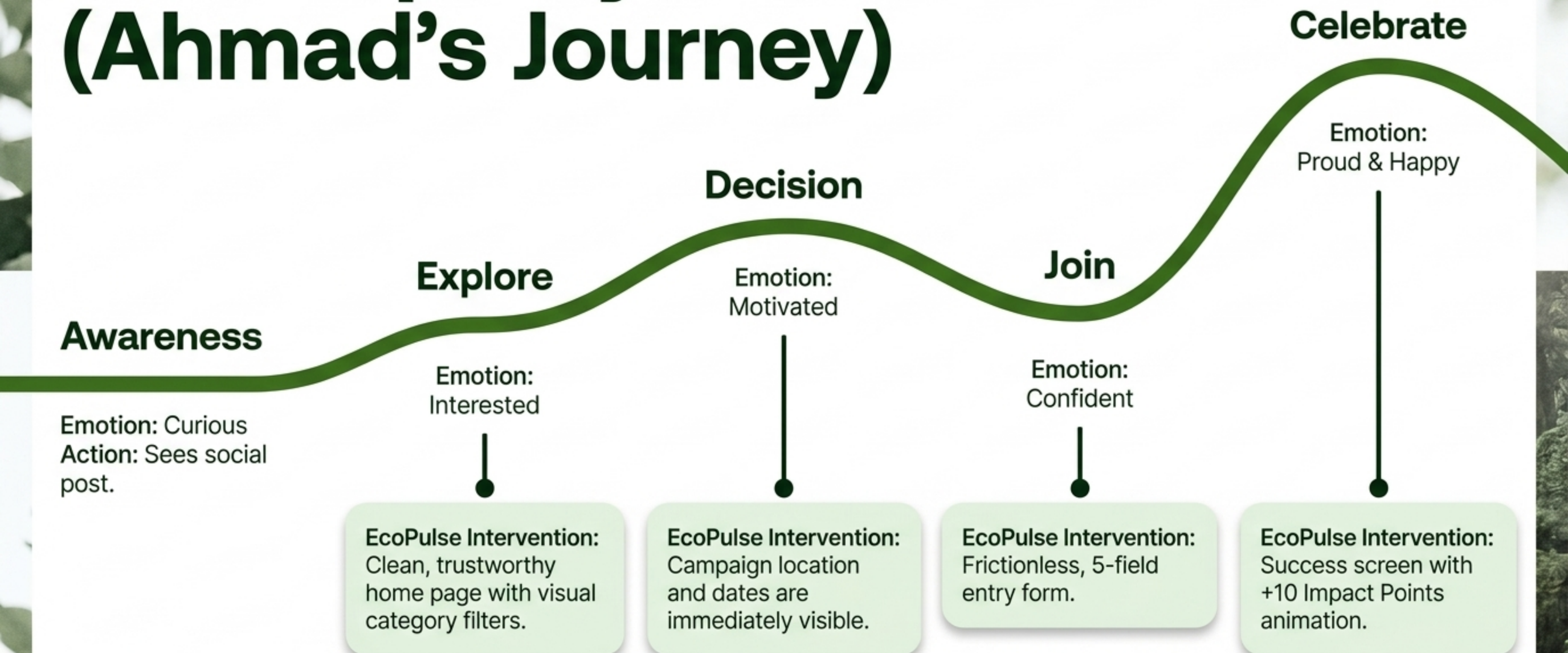
Goals

- Join local campaigns effortlessly
- Track personal environmental impact
- Earn recognition for community contributions

Frustrations

- Hard-to-navigate websites
- Long/confusing sign-up forms
- Radio silence after registration

The Empathy Arc (Ahmad's Journey)



Information Architecture & Task Flow



The architecture is built for velocity. A user can go from platform discovery to confirmed participation in under 3 steps.

Design System & Brand DNA

Colour



Primary: #1B6B35.
Represents nature, trust, and growth. Used for primary actions.

Accent: #D6EDD9.
Used for cards and highlights.



Background: #FFFFFF.
For expansive negative space.

Text: #1B4332.
Deep contrast for accessibility.

Typography

Massive, bold headers to capture short attention spans

Highly legible non-serif for 14-16px readable body copy

Imagery Rule

Emotional anchor. Use real photos of volunteers and nature to connect digital actions to physical results.

Components

Join Now

Volunteer

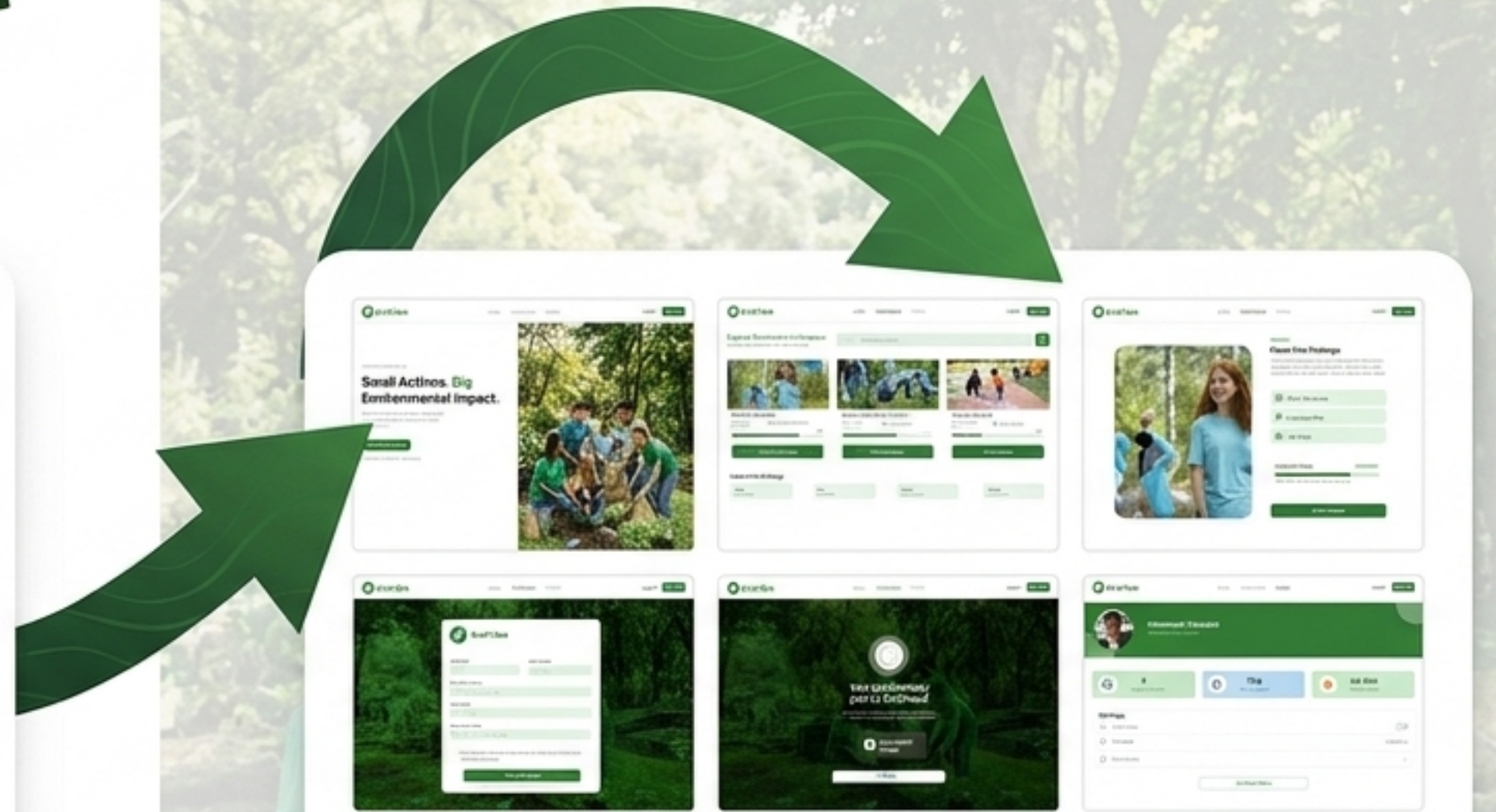


From Blueprint to Reality



Low-Fidelity Architecture.

Focus on element placement, hierarchy, and navigation flow without visual bias.



High-Fidelity Execution.

Injection of brand DNA, emotional imagery, and accessible contrast ratios.

Anatomy of the Desktop Experience

Impact Visualisation:

Immediate display of community stats (10,000+ Members, 5 Tons Reduced) builds instant credibility.



Affordance by Design:

Campaign cards mimic real-world event posters, leveraging existing mental models.

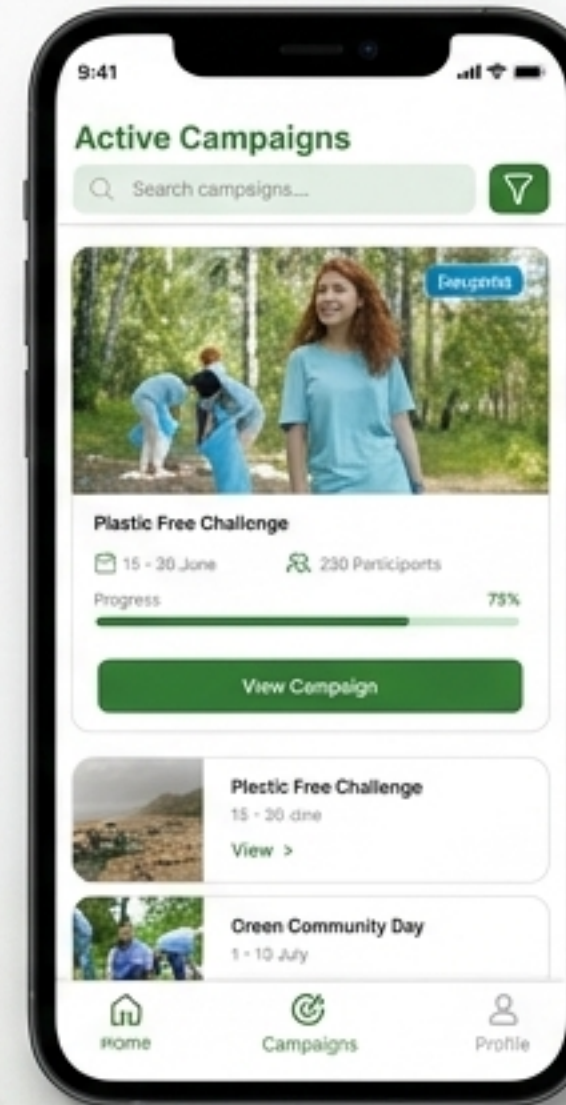
The Urgency Metric:

Progress bars (e.g., 75% collected) provide visual momentum and motivate users to push the campaign to its goal.

The Mobile Translation (Responsive UX) Screen



Formatting: Content elegantly reflows into a single-column, vertical scroll layout suited for one-handed use.

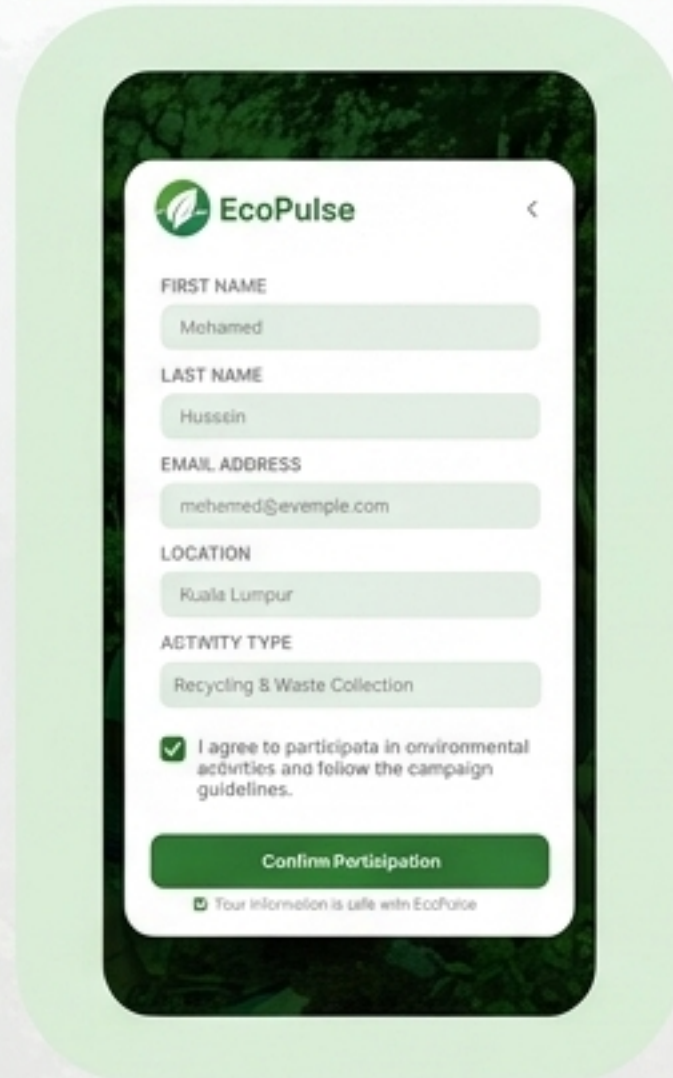


Navigation: Top navigation is replaced by a persistent Bottom Tab Bar (Home, Campaigns, Profile), matching familiar mobile mental models.



Ergonomics: Full-width 'Join Campaign' buttons and expanded form fields accommodate thumb-tap targets, reducing motor-skill friction.

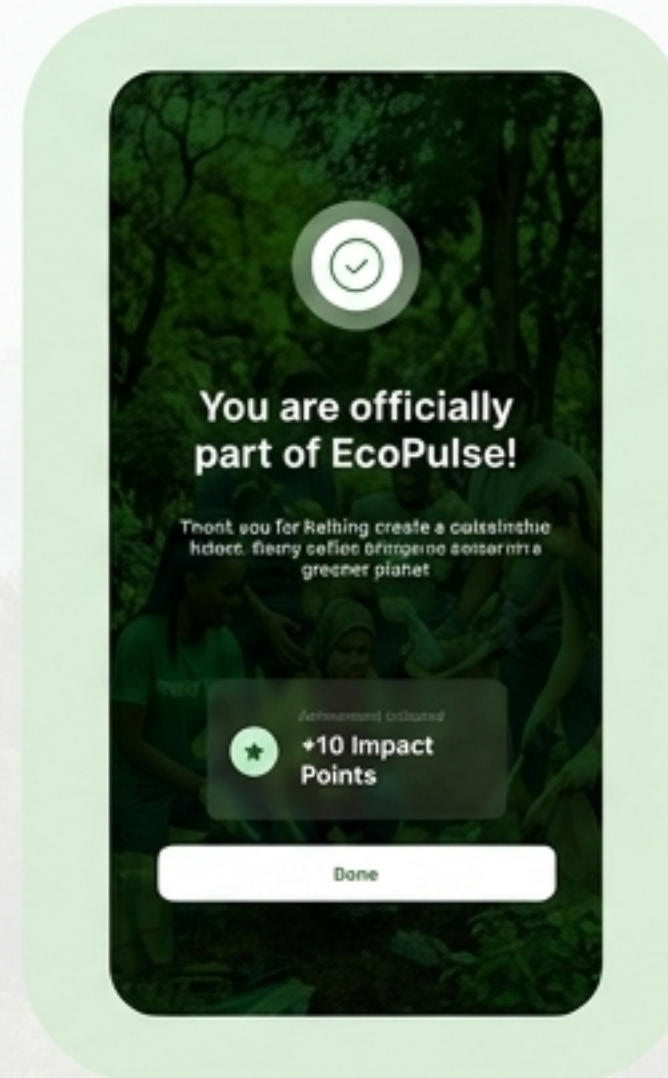
The Psychology of Gamification (Join Flow)



The image shows a mobile app registration form for 'EcoPulse'. The form is titled 'EcoPulse' with a leaf icon. It contains five input fields: 'FIRST NAME' (filled with 'Mohamed'), 'LAST NAME' (filled with 'Hussin'), 'EMAIL ADDRESS' (filled with 'mehemed@evemple.com'), 'LOCATION' (filled with 'Kuala Lumpur'), and 'ACTIVITY TYPE' (filled with 'Recycling & Waste Collection'). Below these fields is a checkbox with the text 'I agree to participate in environmental activities and follow the campaign guidelines.' which is checked. At the bottom is a green button labeled 'Confirm Participation' and a small disclaimer 'Your information is safe with EcoPulse'.

Step 1: Frictionless Entry.

A ruthlessly edited 5-field form. Lengthy T&Cs replaced with a simple checkbox to prevent user drop-off.



Step 2: Immediate Reward.

Satisfies Nielsen's Heuristic #1 (Visibility of System Status). The user is instantly awarded '+10 Impact Points' with a celebratory animation, transforming a functional task into an emotional win.

Usability Evaluation Matrix

Strong Passes

- ✓ **Visibility of System Status:** Instant point rewards and progress bars.
- ✓ **Match with Real World:** Familiar icons (leaf, recycle) and poster-like cards.
- ✓ **Consistency:** Green palette and fonts unified across devices.
- ✓ **Aesthetic Design:** Generous whitespace and minimalist data display.

Actionable Warnings

- ⚠ **User Control:** No visible 'cancel' or 'back' button on the join form.
- ⚠ **Error Prevention & Recovery:** Lack of live validation or red borders for empty fields.
- ⚠ **Help/Documentation:** Absence of an FAQ or tooltips for clarity.

Actionable Iterations (Before vs. Improved)

Current Flaw

User Control: Trapped in the form without an exit path.



Proposed Structural Fix

Add a persistent 'Cancel / X' button with a confirmation dialogue to restore user freedom.

Error Handling: Blank fields submit silently, causing data failure.



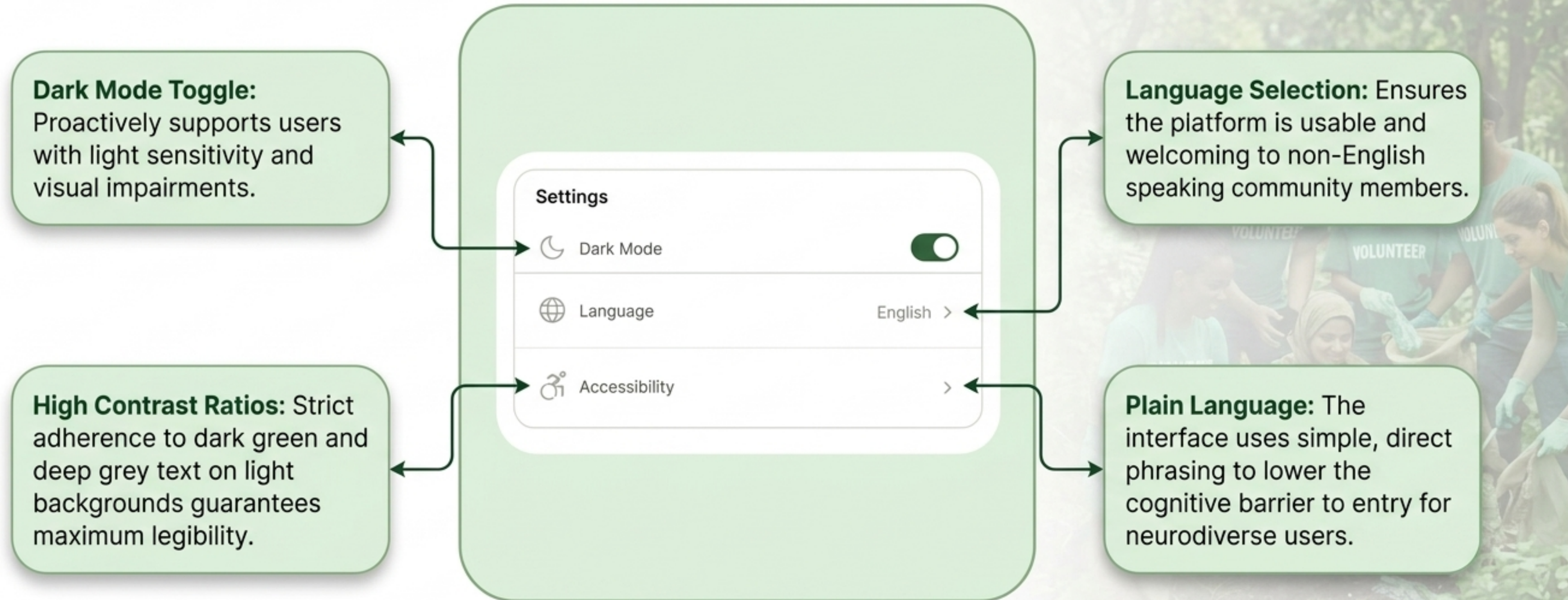
Introduce live red/green border validation and 'This field is required' micro-copy directly under inputs.

Cognitive Load: 'Activity Type' is an open, vague text field requiring guesswork.



Replace with a dropdown menu featuring explicit presets (Recycling, Tree Planting) to reduce cognitive friction.

Accessibility by Design



Professional Reflection & The Roadmap

Lessons Learned

- ✓ True UX is about removing obstacles, not just adding colours.
- ✓ Balancing minimalist aesthetics with necessary information density is an ongoing challenge.
- ✓ Responsive design must be conceptualised early to ensure fluid translation between varied viewports.

The Product Roadmap

- Implement real-time form validation algorithms.
- Integrate a push notification system for upcoming campaign dates.
- Develop social sharing tools to amplify gamification achievements.
- Conduct live user testing cohorts (5+ participants) for further iteration.